Change mass while net force stays fixed

LEARNING TARGET · S3 I can test how mass affects acceleration at fixed force.

A · KEY VOCABULARY

Mass M	All moving mass, including any hanger.
Control	A value held fixed for a fair comparison.
Inverse	Double M halves a at fixed net force.
Slope	Plot a against $1/M$ to get an ideal line.

B · KEY EQUATIONS + UNITS

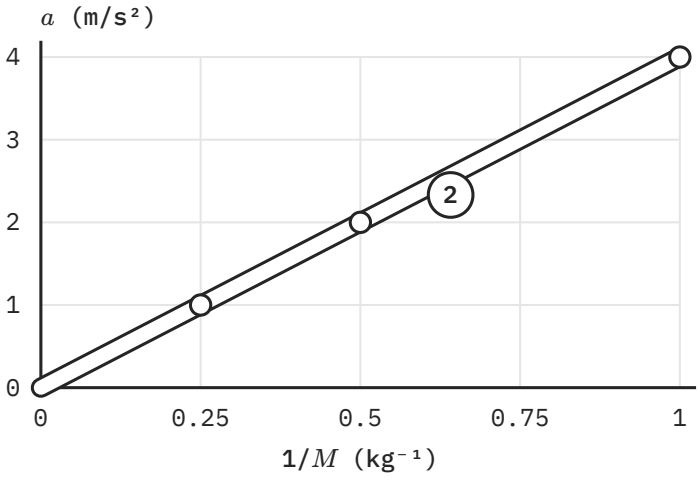
$$a = F_{\text{net}} \div M$$

$$M \times a = F_{\text{net}}$$

$1/M$ inverse mass, kg^{-1}
slope units: N

C · COLOR IT · CHANGE ONLY MASS

*Ideal data; net force **4 N**.*



	M (kg)	a (m/s²)
1	1	4
	2	2
	4	1

3 · F_{net} = 4 N

☐ **1 · PENCIL** Mass and data points

☐ **2 · GREEN** Speeding up: moving +

☐ **3 · ORANGE** Fixed net force

D · WORKED EXAMPLE

Under the same net force, a **1 kg** system accelerates at **4 m/s²** and a **2 kg** system at **2 m/s²**. **Predict a for a 3 kg system.**

GIVEN	EQUATION	WORK	ANSWER
M_1 = **1 kg**; a_1 = **4 m/s²** M_2 = **2 kg**; a_2 = **2 m/s²** same net force	Check $M \times a$. Then $a = F_{\text{net}}/M$.	1(4) = 2(2) = **4 N** a_3 = **4 N** ÷ **3 kg**	**1.33 m/s²** Assumes net force stays **4 N**.

E · THE 2026-XJ THREAD

DAY 11 OF 22 · NEW TOOL: MASS COMPARISON

For the same force, a more massive XJ changes velocity less quickly. The comparison needs the same net force; a larger gravitational force can change that result.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else